

Week 3: Daily Journals

Monday:

Today, I began refactoring our Mode Manager program to improve its architecture and maintainability as we await the switchboard's delivery. I transitioned the codebase to an object-oriented structure, encapsulating each operational mode into its own individual class. Transitioning from a single 1,000-line file to modular 100-line classes will significantly streamline debugging and future development. Because physical hardware was unavailable today, all development and syntax verification were conducted locally on the Ubuntu host laptop.

Tuesday:

I continued modularizing the Mode Manager program. Today, I integrated the Raspberry Pi 4 diagnostic test routine developed in previous weeks directly into the Idle mode class. This integration eliminates the need to execute a separate standalone script during FlatSat testing. Due to ongoing hardware limitations, development and initial code validation were performed on the Ubuntu environment.

Wednesday:

Development on the Mode Manager program continued with the integration of the 630 nm camera software developed last semester. I embedded this legacy code directly into the 630 nm Camera mode class to prepare the system for upcoming optical testing. Work remained restricted to local simulation on the Ubuntu laptop due to a lack of physical access to the Raspberry Pi 4.

Thursday:

I focused today's development on integrating Kevin's Software-Defined Radio (SDR) data-collection codebase. This script was successfully implemented into the HF Receiver mode class to enable automated SDR testing within the main execution loop. As with previous days, development was completed on the Ubuntu environment.

Friday:

With physical hardware access restored, I conducted integration testing of the newly refactored Mode Manager, specifically the Pi 4 diagnostic routine, on the Raspberry Pi 4. I encountered a few runtime exceptions during initial execution; however, after systematic troubleshooting and debugging, the code performed exactly as expected. Next week, I plan to continue testing the remaining operational modes, so the software is fully validated by the time the switchboard arrives for complete FlatSat testing.

Pictures



Figure 1: Jonah Programming